

BRANDON KONG

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EDUCATION

University of Waterloo

Honours Bachelor of Computer Engineering

Sept 2023 - Present

Waterloo, ON

- Relevant Coursework: Algorithms and Data Structures, Systems Programming, Concurrency, Compilers, Real-Time Operating Systems

EXPERIENCE

Royal Bank of Canada (Global Risk Management)

AI Intern

May 2026 – Present

Toronto, ON

- Developing a GenAI evaluation system to perform **gap assessment** on model whitepapers against internal standards

WAT.ai

Machine Learning Engineer

Jan 2026 – Present

Waterloo, ON

- Built a Python, PostgreSQL, and FAISS data ingestion pipeline for regulatory approval workflows at Bindwell (YC W25), processing **470+** HTML documents.

Rogers Communications

AI / GenAI Intern

Sept 2025 – Dec 2025

Toronto, ON

- Spearheaded a bundle **recommendation engine** with AWS Sagemaker, increasing user conversion rate by **30%** and projected to save over **\$50M** annually
- Built agent orchestration layer with AWS Bedrock + Lex, reducing interaction time by **80%** across user workflows.
- Designed an **ETL + NLP** processing pipeline to analyze and extract insights from **20,000+** employee complaints
- Implemented **LLM guardrails** to enforce policy compliance and prevent unsafe user queries.

Kisoji Biotechnology

Machine Learning Engineer Intern

May 2025 – Sept 2025

Toronto, ON

- Developed and trained models in Pytorch for antibody generation, improving candidate screening efficiency by **20%**
- Improved PyTorch training throughput by **30%** using CUDA-based GPU acceleration and kernel tuning

Adanomad

Software Engineer Intern

Jan 2025 – Apr 2025

Hamilton, ON

- Built workflow engine with **LRU** caching and parallel execution for multi-step pipelines, reducing latency by **50%**.
- Scaled real-time voice agent system to **200+** concurrent users in production environments.

PROJECTS

ClassiMail | Python, Pytorch, scikit-learn, OpenAI, Docker

- Built a email retrieval and ranking system to automatically recommend relevant emails for users
- Leveraged a **CTR-based** ranking algorithm using 8 weighted features, improving email click-through rates by **35%**
- Built an ML ranking pipeline with **feature engineering** for 15+ signals, achieving **0.75 AUC** on click prediction
- Developed hybrid vector search with FAISS and OpenAI embeddings, boosting keyword matches by **1.5x**

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL, HTML, CSS, Bash

Frameworks: Pytorch, LangChain, Tensorflow, scikit-learn, NumPy, React, Next.js, FastAPI, Express.js

Tools: AWS, Docker, Git, Postgres, DynamoDB, Supabase, REST APIs, GraphQL, WebSockets, FAISS